

STAT 709: Homework 3

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Fall 2026

1. Textbook exercises

Jun Shao's book, Chapter 1: Exercises 50 and 53 (required); Exercise 91 (optional).

2. Textbook exercises

Durrett's *Probability: Theory and Examples*, fifth edition: Exercises 2.1.4 and 2.1.6.

The remaining exercises concern conditional expectation and are optional. Read the questions and try as many parts as you can.

3. Conditioning on a transformation (optional)

Let $X \in \mathbb{R}^n$ have a continuous density p , and let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuous, with $\mathbb{E}\|f(X)\|_2 < \infty$. Let $A_1, \dots, A_m$ be pairwise disjoint open sets whose union has a Lebesgue-null complement. On each A_j , assume that g is continuously differentiable and one-to-one, with $\det Dg(x) \neq 0$. Write $V_j = g(A_j)$ and let $h_j : V_j \rightarrow A_j$ be its inverse.

We want a formula for $\mathbb{E}[f(X) \mid g(X) = y]$. In general, $(f(X), g(X))$ does not have a joint density on $\mathbb{R}^{2n}$, so the elementary joint-density ratio formula need not apply directly.

For the following local calculation, fix $y \notin \bigcup_{j=1}^m \partial V_j$. Put $J(y) = \{j : y \in V_j\}$, and assume that $p(h_j(y)) > 0$ for at least one $j \in J(y)$. These conditions ensure that a sufficiently small cube around y meets exactly the active inverse branches and that the limiting denominator is positive. For $\Delta > 0$, let $E_\Delta = \prod_{i=1}^n (y_i - \Delta, y_i + \Delta)$.

(1) For $j \in J(y)$ and sufficiently small Δ , define $B_\Delta^{(j)} = h_j(E_\Delta)$. Prove that

$$\text{Vol}(B_\Delta^{(j)}) = |\det Dh_j(y)| (2\Delta)^n + o(\Delta^n) \quad (\Delta \downarrow 0).$$

(2) Use part (1) to evaluate

$$\lim_{\Delta \downarrow 0} \frac{\mathbb{E}[f(X) \mathbf{1}\{g(X) \in E_\Delta\}]}{\mathbb{P}(g(X) \in E_\Delta)}.$$

Derive a version of $\mathbb{E}[f(X) \mid g(X) = y]$ in terms of the inverse branches. Explain why this formula, summed over the branches defined at y , gives a conditional expectation for almost every y under the law of $g(X)$.

4. Conditional independence (optional)

Let X, Y, Z be real-valued random variables. Assume that X and Y are conditionally independent given Z , in the sense that

$$\mathbb{P}(B \mid X, Z) = \mathbb{P}(B \mid Z) \quad \text{a.s. for every } B \in \sigma(Y).$$

The first two parts connect this definition with the familiar density formulation.

- (1) Let (U, W) be random vectors with a joint density. For a Borel function φ satisfying $\mathbb{E}|\varphi(U)| < \infty$, show that a version is

$$\mathbb{E}[\varphi(U) \mid W = w] = \int \varphi(u)p(u \mid w) du$$

for almost every w under the law of W , where $p(u \mid w) = p_{U,W}(u, w)/p_W(w)$ when $p_W(w) > 0$.

- (2) If (X, Y, Z) has a joint density, show that

$$f_{X,Y|Z}(x, y \mid z) = f_{X|Z}(x \mid z)f_{Y|Z}(y \mid z)$$

for almost every z under the law of Z and Lebesgue-almost every (x, y) .

- (3) Without assuming a joint density, prove that, for every $A \in \sigma(X)$ and $B \in \sigma(Y)$,

$$\mathbb{P}(A \cap B \mid Z) = \mathbb{P}(A \mid Z)\mathbb{P}(B \mid Z) \quad \text{a.s.}$$

Hint: Use the tower property and the definition of conditional expectation.

- (4) Let $\mathbf{X} = (X_1, \dots, X_p)^\top \sim N_p(0, \Sigma)$, with Σ positive definite, and define the precision matrix $\Theta = \Sigma^{-1}$. Set $Y = X_1$ and $Z = (X_2, \dots, X_p)^\top$. In the coordinate order (Y, Z) , write

$$\Sigma = \begin{pmatrix} \sigma_{YY} & \sigma_{ZY}^\top \\ \sigma_{ZY} & \Sigma_{ZZ} \end{pmatrix}.$$

Recall that

$$Y \mid Z = z \sim N(z^\top \Sigma_{ZZ}^{-1} \sigma_{ZY}, \sigma_{YY} - \sigma_{ZY}^\top \Sigma_{ZZ}^{-1} \sigma_{ZY}), \quad \theta_{ZY} = -\sigma_{ZY}^\top \Sigma_{ZZ}^{-1} \sigma_{ZY}.$$

In a Gaussian graphical model, distinct nodes are joined when the corresponding precision-matrix entry is *nonzero*; a zero entry indicates a missing edge. Prove that $\theta_{ij} = 0$ implies that X_i and X_j are conditionally independent given all the other coordinates.